Project 1 Creating Database (150 Points)

Instructions:

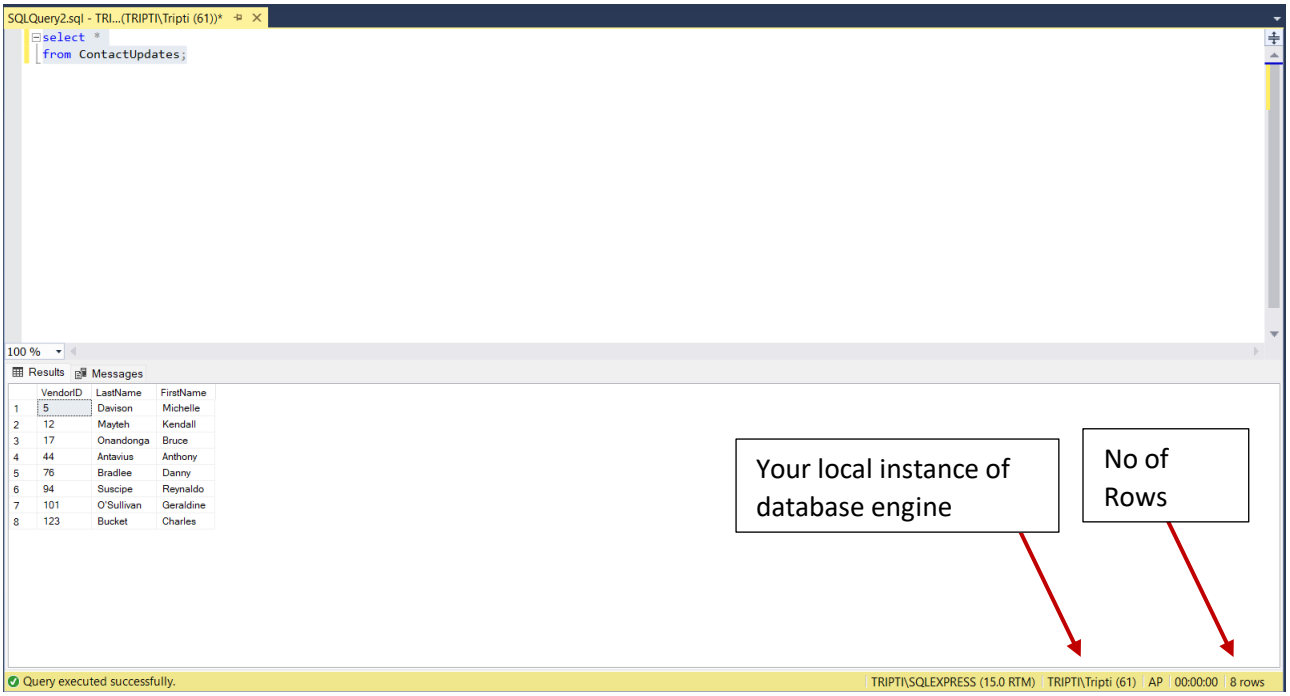
Based on the following (after sample screenshot) ER diagram, relational schema, create HAFH Database. Then, Insert the sample data into the database. After inserting the data, please take a screenshot for each table. The screenshot format requirements are listed at the end of the assignment.

Deliverables:

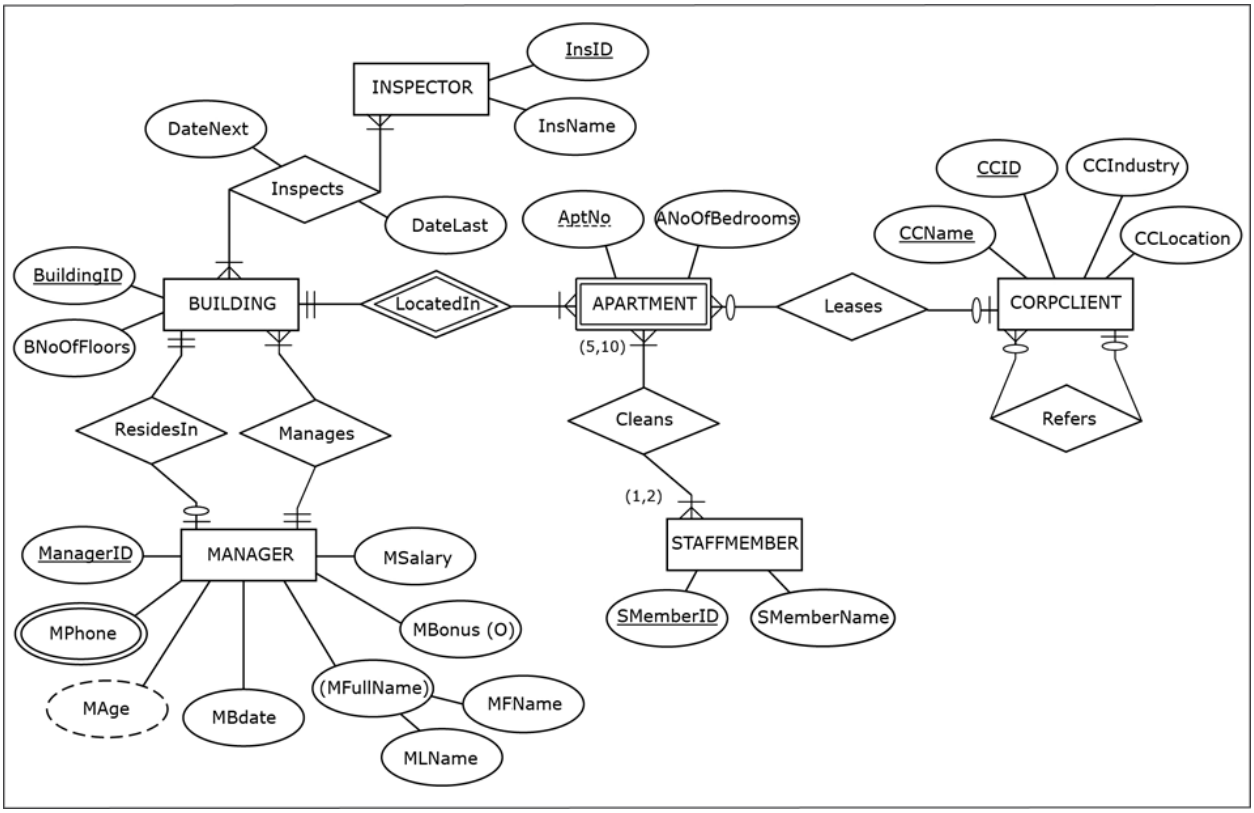
1. All Sql queries for the HAFH reality database. This includes queries for creating the database, tables data loading (insert query), and then the queries showing the content of every table (e.g., show table Manager, the number of columns it has, and the instances showing the data, i.e., number of rows and columns in the table-see sample screenshot) (30 points)
2. Make sure to label each part of the assignment clearly. A lack of clarity in the submission (including SQL codes and screenshots) may result in significant points deduction.
3. Screenshots after running SQL query to extract all the columns from tables. There are nine tables, so there should be nine screenshots. Requirements for screenshots:
    a. One screenshot for one table (5*9 = 45 points)
    b. Show all columns in the table (5*9 = 45 points)
    c. The bottom of the screenshot shall show the following information:
        - Your local instance of the database engine; and
        - Number of rows in the table. (note: if a table has many rows that cannot be displayed in one screen, you don't need to show those rows that are not displayed in the first page. Just take the screenshot of the first page and show the number of rows at the bottom of the screenshot. That is good enough).
4. Display the total amount HAFH spends on manager salaries (as TotalSalary) and the total amount HAFH spends on manager bonuses (as TotalBonus). 5 Points
5. Display the ManagerID, MFName, MLName, MSalary, and MBonus, for all managers with a salary greater than $50,000 and a bonus greater than $1,000. 5 Points
6. Display the BuildingID, BNoOfFloors, and the manager's MFName and MLName for all buildings. 5 Points
7. Display the BuildingID and AptNo for all apartments leased by the corporate client WindyCT 5 Points
8. Display the InsID and InsName for all inspectors that have any inspections scheduled after 1-APR-2020. Do not display the same information more than once. 5 Points
9. Display the ManagerID, MFName, MLName, MSalary, and MBonus for the manager with the lowest total compensation (defined as salary plus bonus) 5 Points.

Provide all sql queries and the screenshots.

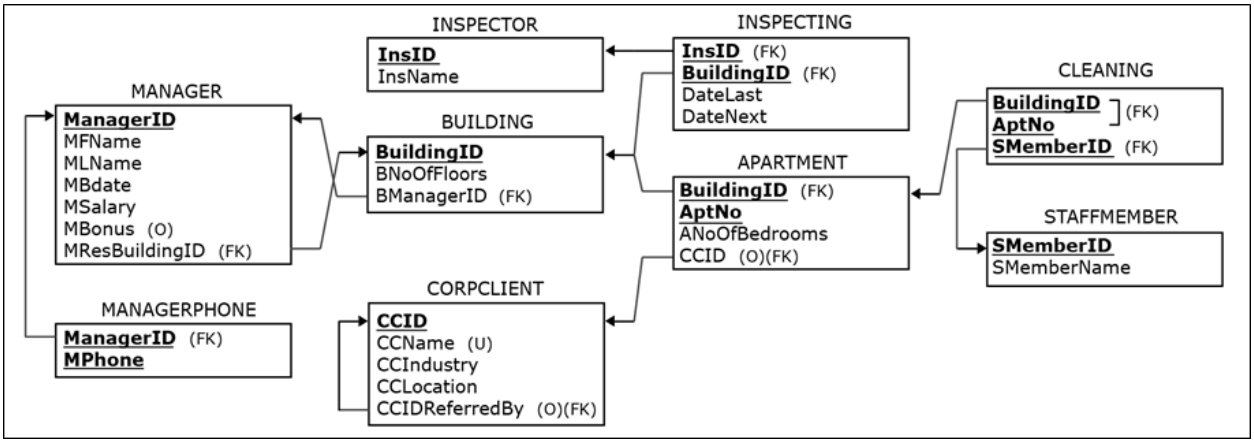
Sample screenshot:



ER Diagram and Relational Schema- HAFH Reality



In the manager table don't list MResBuildingID as a FK. Create it as a regular column here.



Sample Data Record

INSPECTOR

InsID	InsName
I11	Jane
I22	Niko
I33	Mick

BUILDING

BuildingID	BNoOfFloors	BManagerID
B1	5	M12
B2	6	M23
B3	4	M23
B4	4	M34

APARTMENT

BuildingID	AptNo	ANoOfBedrooms	CCID
B1	41	1	
B1	21	1	C111
B2	11	2	C222
B2	31	2	
B3	11	2	C777
B4	11	2	C777

INSPECTING

InsID	BuildingID	DateLast	DateNext
I11	B1	5.5.2019	5.5.2020
I11	B2	2.2.2019	2.2.2020
I22	B2	3.4.2019	2.2.2020
I22	B3	9.8.2019	3.3.2020
I33	B3	4.4.2019	4.4.2020
I33	B4	5.5.2019	4.4.2020

CLEANING

BuildingID	AptNo	SMemberID
B1	21	5432
B1	41	9876
B2	11	9876
B2	31	5432
B3	11	5432
B4	11	7652

MANAGER

ManagerID	MFName	MLName	MBDate	MSalary	MBonus	MResBuildingID
M12	Boris	Grant	6.6.1988	60000		B1
M23	Austin	Lee	2.2.1983	50000	5000	B2
M34	George	Sherman	7.8.1984	52000	2000	B4

MANAGERPHONE

ManagerID	MPhone
M12	555-2222
M12	555-3232
M23	555-9988
M34	555-9999

STAFFMEMBER

SMemberID	SMemberName
5432	Brian
9876	Boris
7652	Caroline

CORPCLIENT

CCID	CCName	CCIndustry	CCLocation	CCIDReferredBy
C111	BlingNotes	Music	Chicago	
C222	SkyJet	Airline	Oak Park	C111
C777	WindyCT	Music	Chicago	C222
C888	SouthAlps	Sports	Rosemont	C777

Data types

Column Name	Data Type	Column Name	Data Type	Column Name	Data Type
Managerid	Char(4)	**Buildingid**	Char(3)	**Ccname**	Varchar(25)
Mfname	Varchar(15)	**Bnooffloors**	INT	**Ccindustry**	Varchar(25)
Mlname	Varchar(15)	**Bmanagerid**	Char(4)	**Cclocation**	Varchar(25)
Msalary	Numeric(9,2)	**Insid**	Char(3)	**Ccidreferredbby**	Char(4)
Mbdate	Date	**Insname**	Varchar(15)	**Aptno**	Char(5)
Mbonus	Numeric(9,2)	**Datelast**	Date	**Anoofbedrooms**	INT
Mresbuildingid	Char(3)	**Datenext**	Date	**Smemberid**	Char(4)
Mphone	Char(11)	**Ccid**	Char(4)	**Smembername**	Varchar(15)

Quick notes:

The error that you may notice when creating a cleaning table:

You will get a message as shown in red if you code the cleaning table in the manner it is shown in the first query. The error message is happening because the apartment table has a composite primary key (buildingID, AptNo). That's why when you try to reference AptNo as a single

primary key as I mentioned in the top query, SQL can not find it as ‘unique’ primary key in the apartment table. Because apartment table has a composite primary key, it should be referenced as the composite primary key in the cleaning table.

```
create table cleaning
(buildingID char (3),
AptNo char (5),
SMemberID char (4),
primary key (buildingID, AptNo, SmemberID),
foreign key (buildingID) references building (buildingID),
foreign key (AptNo) references Apartment (AptNo),
foreign key (SMemberID) references staffmember (SMemberID));

create table cleaning1
(buildingID char (3),
AptNo char (5),
SMemberID char (4),
primary key (buildingID, AptNo, SmemberID),
foreign key (buildingID, AptNo) references Apartment (buildingID, AptNo),
foreign key (SMemberID) references staffmember (SMemberID));
```

%

Messages

```
Msg 1776, Level 16, State 0, Line 74
There are no primary or candidate keys in the referenced table 'Apartment' that match the referencing column list in the fo
Msg 1750, Level 16, State 1, Line 74
Could not create constraint or index. See previous errors.
```